Math Inquirers

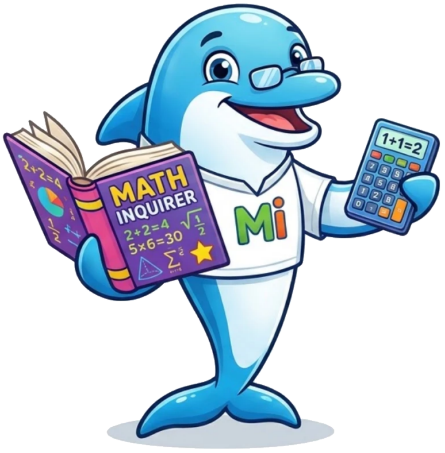
Think. Question. Prove.

Level 1 (ages 6-8) - Term 2

Home Practice Pack

Book 3

Lessons 11 to 15



One sheet for each lesson. Tear out a sheet and do it at home.

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Who Is Where?: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use toys in a row, or draw the line. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

Put 6 toys in a row. Put a coin on the 4th toy from the front. The coin is on the toy that is _____ from the back.

No.2

Start here

Kumar is 7th in the line. There are _____ children in front of Kumar.

No.3

Start here

Aisha is 2nd. Wei is right behind Aisha. Lily is right behind Wei. Lily is _____ in the line.

No.4

Think harder

Jenny is 4th from the front and 4th from the back. How many children are in the line? Write: I know because ...

No.5

Go further

Draw a line of 9 children. Ali is in the middle. Ben stands 2 places behind Ali. Ben is _____ from the front.

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 3rd
From the front the coin is on the 4th toy. Count from the back: the last toy is 1st, then 2nd, then the coin toy is 3rd.

2. 6 children
The children in front are the 1st to the 6th. That is 6 children. Kumar himself is the 7th.

3. 4th
Wei is right behind Aisha, so Wei is 3rd. Lily is right behind Wei, so Lily is 4th.

4. 7 children
In front of Jenny are 3 children, behind her are 3 children, and Jenny makes 1. 3 + 1 + 3 = 7. It is not 8, because 4 + 4 counts Jenny twice.

5. 7th
A line of 9 has 4 children in front of the middle child and 4 behind, so Ali is 5th. Ben is 2 places behind Ali, so Ben is 5 + 2 = 7th.

More practice online (no timers, no scores): ask your tutor for the practice link.

Bar Strips: More and Fewer: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use paper strips or draw boxes in a row. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

Ali has 14 pencils. Mei has 9 pencils. Draw a strip for each. Ali has _____ more pencils than Mei.

No.2

Start here

Ravi has 38 marbles. Kumar has 20 more marbles than Ravi. How many marbles does Kumar have?

No.3

Start here

Siti has 25 stickers. Wei has 15 more stickers than Siti. How many stickers do they have altogether?

No.4

Think harder

Ali and Ben have 30 sweets altogether. Ali has 6 more sweets than Ben. How many sweets does Ben have? Write: I know because ...

No.5

Go further

Mei has 24 stickers. Lily has 10 stickers. Mei gives some stickers to Lily. Then they have the same number. How many stickers does Mei give?

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 5 more pencils
Ali's strip is longer. The gap is 14 take away 9, which is 5.

2. 58 marbles
Kumar has Ravi's 38 and 20 more: 38 + 20 = 58.

3. 65 stickers
Wei has 25 + 15 = 40 stickers. Altogether: 25 + 40 = 65.

4. 12 sweets
Take the extra 6 away from 30: 24 is left. Two equal strips share 24: 12 each. Ben has 12 and Ali has 18. Check: 12 + 18 = 30.

5. 7 stickers
Together they have 34, so each has 17 at the end. Mei goes from 24 to 17, so she gives 7. Half of the gap of 14 is 7.

More practice online (no timers, no scores): ask your tutor for the practice link.

Doubles and Near Doubles: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use counters or draw. Small things stay on the table, away from mouths and little brothers and sisters. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

Fill in the missing numbers. (a) Double 4 is ☐. (b) Double 6 is ☐☐. (c) Double 8 is ☐☐.

No.2

Start here

Mei has 7 red marbles and 7 blue marbles. How many marbles does she have altogether? She has _____ marbles.

No.3

Start here

Fill in the missing numbers. (a) Double 20 is ☐☐. (b) Double 40 is ☐☐.

No.4

Think harder

Add. (a) 6 + 7 = ☐☐ (b) 9 + 10 = ☐☐

No.5

Go further

Add 7 + 8 + 7 + 6 + 7. The total is _____.

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. **8, 12, 16**
4 + 4 = 8. 6 + 6 = 12. 8 + 8 = 16.

2. **14**
7 + 7 = 14. Two of the same number is a double.

3. **40, 80**
Double 2 tens is 4 tens: 40. Double 4 tens is 8 tens: 80.

4. **13, 19**
Double 6 is 12 and 1 more is 13. Double 9 is 18 and 1 more is 19.

5. **35**
The 8 is 1 more than 7 and the 6 is 1 less, so the sum is five 7s: 35.

More practice online (no timers, no scores): ask your tutor for the practice link.

Odd + Odd: Always, Sometimes, Never: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use counters or draw dots in pairs. Small things stay on the table, away from mouths and little brothers and sisters. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

Pair up counters. Write Odd or Even. (a) 8 _____ (b) 15 _____ (c) 20 _____

No.2

Start here

Ali has 13 counters. He puts them in pairs. How many counters are left over?

_____ counter is left over.

No.3

Start here

Fill in the missing numbers. The last digits of even numbers are 0, 2, 4, ☐ and ☐.

No.4

Think harder

Add. Circle Odd or Even for the total. (a) 5 + 6 = _____ Odd or Even (b) 7 + 9 = _____ Odd or Even

No.5

Go further

Write two odd numbers that add to 20. Write two odd numbers that add to 30. Many answers are right. _____ + _____ = 20 _____ + _____ = 30

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. Even, Odd, Even
8 and 20 make pairs with none left over: even. 15 leaves one over: odd.

2. 1
13 counters make 6 pairs and 1 is left over.

3. 6, 8
Even numbers end in 0, 2, 4, 6 or 8.

4. 11 odd; 16 even
5 + 6 is odd + even, so odd (11). 7 + 9 is odd + odd, so even (16).

5. Any two odd numbers: for example 9 + 11 = 20 and 15 + 15 = 30 (the same number twice is fine)
Odd plus odd gives an even number, so 20 and 30 can be made. Check each pair adds to the total.

More practice online (no timers, no scores): ask your tutor for the practice link.

Equal Groups: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use counters on plates or draw dots. Small things stay on the table, away from mouths and little brothers and sisters. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

Ali puts 3 sweets on each of 4 plates. Fill in the blanks. (a) 3 + 3 + 3 + 3 = ☐☐

(b) 4 groups of 3 = ☐☐

No.2

Start here

Mei has 2 bags with 3 sweets in each bag and 1 bag with 4 sweets. Are the groups equal? Circle Yes or No.

Yes No

No.3

Start here

Fill in the blanks. (a) 5 + 5 + 5 = ☐☐ (b) 3 groups of 5 = ☐☐ (c) 3 × 5 = ☐☐

No.4

Think harder

Is 2 × 6 = 6 × 2? Circle Yes or No. Use 12 counters or draw dots to check.

Yes No

No.5

Go further

A tricycle has 3 wheels. A bike has 2 wheels. How many wheels do 4 tricycles and 3 bikes have in all? _____ wheels

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 12 and 12
4 plates with 3 sweets on each: 3 + 3 + 3 + 3 = 12.

2. No
The bags have 3, 3 and 4 sweets. The groups are not equal.

3. 15, 15, 15
5 + 5 + 5 = 15. 3 groups of 5 and 3 × 5 mean the same: 15. 5 × 3 = 15 is also right.

4. Yes
2 rows of 6 is 12 and 6 rows of 2 is 12, so the totals are the same.

5. 18
4 tricycles: 3 + 3 + 3 + 3 = 12. 3 bikes: 2 + 2 + 2 = 6. 12 + 6 = 18.

More practice online (no timers, no scores): ask your tutor for the practice link.